ClipSAT : IB Diploma Mathematics: Analysis & Approaches / Applications

Unit 1.01: Topic 1: Number & Algebra Sequences, Logs & Binomial

Name: _____

Q1. Consider the function fx = 2x + 5 . What is the value of f3 ?

[ ] A: 11
[ ] B: 8
[ ] C: 10
[ ] D: 16

Q2. The graph of the quadratic function fx = x^2 - 6x + 5 is shown below. Find the value of the y-intercept.

[ ] A: 5
[ ] B: -5
[ ] C: 0
[ ] D: 1

Q3. Consider the function fx = x^2 - 4x + 3 and the line gx = x - 1 . The graph of these functions is shown below. Find the coordinates of the point where the line intersects the parabola.

[ ] A: 1, 0
[ ] B: 3, 2
[ ] C: 2, 1
[ ] D: 4, 3

Q4. e^0=

[ ] A: 0
[ ] B: 1
[ ] C: e
[ ] D: âˆž

Q5. The graph of the quadratic function fx = ax^2 + bx + c is shown below. It passes through the points 0, -3 , 1, 0 , and -1, -4 . Find the value of c .

[ ] A: -3
[ ] B: 3
[ ] C: -1
[ ] D: 1

Q6. Consider the function fx = 2x - 3 . Determine the value of f3 .

[ ] A: 3
[ ] B: 6
[ ] C: 9
[ ] D: 0

Q7. Solve for x : 3^x=9 .

[ ] A: 3
[ ] B: 4
[ ] C: 2
[ ] D: 1

Q8. An arithmetic sequence has first term u_1=7 and common difference d=2 . Find u_{15} .

[ ] A: 37
[ ] B: 35
[ ] C: 33
[ ] D: -21